

TEAM CONTRIBUTION & WORK DISTRIBUTION REPORT

Course: DLE-AI-202 (Deep Learning), Cohort I 2026 | Track 1: Pure Research

Project: Do Register Tokens Regularize Vision Transformers Under Data Scarcity? A Controlled 12-Run Ablation

Repository: github.com/shahin1717/vit-research | Submission Date: September 8, 2026

1. Executive Overview & Compliance Statement (§6)

Per Section 6 of the project specifications, this report documents the workload allocation, code ownership, and technical deliverables across our team. The project investigated the regularizing capacity of register tokens ($K \in \{0, 1, 4, 8\}$) under extreme data scarcity (~10k CIFAR-100 images). All five members contributed equally (20.0% per member) across theoretical formulation, PyTorch engineering, A100 GPU sweeps, statistical reduction, and manuscript drafting.

2. Responsibility Matrix & Effort Allocation

Member Name	GitHub / Contact	Core Ownership Domain	Primary Assigned Codebase Modules	Effort
Shahin Alakparov	shahin1717 sahinalekperov5@gmail.com	Core Architecture & Training Lead	src/models/register_vit.py, scripts/train.py, scripts/eval.py	20.0%
Gulnisa Abdurahmanli	gulnisa gulnisa.abdurahmanli@gmail.com	Data Engineering & Multi-Budget Lead	src/data/cifar100_subset.py (100 & 300 imgs/class), src/data/__init__.py	20.0%
Narmina Ibrahimova	nnrmina narminaibrahimova2@gmail.com	Metrics & Interpretability Lead	src/models/attention_hook.py, src/metrics/entropy.py, outliers.py	20.0%
Emil Ahmedli	emilahmedli5 emilahmedli1905@gmail.com	Ablation Sweeps & Hardware Lead	scripts/run_sweep.sh, scripts/run_databudget_sweep.sh, configs/	20.0%
Rufet Dosteliyev	rufetdosteliyev rufetdosteliyev@gmail.com	Ablation Analytics, Figures & Paper Lead	src/utlis/aggregate_databudget.py, scripts/plot_*.py, paper/	20.0%
TOTALS	5 Team Members	Balanced End-to-End Pure Research	24/24 Ablation Runs Verified (100% Passing Tests)	100.0%

3. Detailed Technical Contributions per Member

- Shahin Alakparov (20.0%):** Engineered RegisterVisionTransformer wrapping timm ViT-Tiny ($d=192$). Implemented prepend logic for learnable registers $R \in \mathbb{R}^{K \times d}$ and output slicing. Developed the AMP training harness in scripts/train.py (AdamW, linear warmup + cosine annealing, gradient clipping, checkpoint serialization) and verified 59/59 tests.
- Gulnisa Abdurahmanli (20.0%):** Built generalized StratifiedCIFAR100Subset in src/data/cifar100_subset.py, sampling both 100 and 300 samples/class (10k and 30k images) with deterministic generator seeds. Implemented balanced 9k/1k and 27k/3k train/val splits, Bicubic RandomResizedCrop(224), AutoAugment, and standard test set evaluation loader with pinned memory and multi-worker prefetching.
- Narmina Ibrahimova (20.0%):** Authored ViTAttentionHookManager in src/models/attention_hook.py, non-invasively intercepting attention weights across all 12 MHSA layers with automatic CUDA memory clearing. Coded Shannon attention entropy $H(A^{(l)})$ in src/metrics/entropy.py, patch-norm 3σ outlier rate in src/metrics/outliers.py, and generalization gap metrics.
- Emil Ahmedli (20.0%):** Led execution of all 24 ablation sweep runs across both data budgets ($12 \times 100\text{pc}$ and $12 \times 300\text{pc}$). Configured WireGuard VPN (team1.conf) and remote NVIDIA A100-SXM4 GPU cluster environment. Authored automated execution runners scripts/run_sweep.sh and scripts/run_databudget_sweep.sh with GPU cache clearing, supervising zero-OOM execution across 8+ hours.
- Rufet Dosteliyev (20.0%):** Led multi-seed ablation analytics, statistical reduction, and scientific reporting across both 100pc and 300pc experimental suites. Built aggregation engines (src/utlis/logger.py and src/utlis/aggregate_databudget.py) producing sweep_summary.json and summary.json. Authored comparative visualizers (scripts/plot_metrics.py, scripts/plot_databudget_comparison.py), LaTeX tables, and manuscript Section 5.

4. Academic Integrity Attestation & Sign-Off

We certify that this report accurately reflects the collaborative division of labor for our final project. All members actively reviewed, validated, and approved the codebase, empirical findings, manuscript, and oral defense deck.

Shahin Alakparov Core Architecture Lead Status: Verified & Signed	Gulnisa Abdurahmanli Data Pipeline Lead Status: Verified & Signed	Narmina Ibrahimova Metrics & Hooks Lead Status: Verified & Signed
Emil Ahmedli Ablation Sweeps & Cluster Operations Lead Status: Verified & Signed	Rufet Dosteliyev Ablation Analytics, Figures & Paper Lead Status: Verified & Signed	